

Multisensorial Audiobooks

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Abstract—Electronic books (e-books) and audiobooks are gaining popularity in the last years. Both have been the subject of studies to expand their basic functionalities. In the case of e-books, many studies are found, such as enhanced e-books and Multisensorial Books. On the other hand, little has aimed to extend audiobooks, for example, the interactive audiobooks. In this scenario, our work proposes Multisensorial Audiobooks, a new extension of traditional audiobooks. This model uses multisensorial media elements to enhance the quality of experience of listeners. Multisensorial Audiobooks is created to consider accessibility; that is, it can benefit visually impaired individuals, and, at the same time, dyslexia individuals. We also present a Multisensorial Audiobooks prototype, named as MulSeBook. It is based on the Multisensorial Books model, which has presented promising results in previous studies. Preliminary tests with MulSeBook architecture indicates that the Multisensorial Audiobooks model can be implemented as an extension of traditional audiobooks. This work also presents the first story created to be executed with MulSeBook.

Index Terms—multisensorial, audiobook, multiple sensorial media, MulSeMedia, sensorial

I. INTRODUCTION

In recent years, electronic books (e-books) and audiobooks have been rapidly gaining popularity with widely accessible digital download and streaming services [1]–[4]. In line with this information, reports indicate that Amazon sold more e-books than paper books in 2012 [5] and the Audio Publishers Association estimates that audiobook sales totaled more than \$1.77 billion in 2015 [6]. E-books and audiobooks have received even more notoriety for being used for both leisure and education [7]–[11].

Faced with this scenario, several studies have concentrated on research questions related to e-books and audiobooks [12]–[17]. Another line of studies examines how to enhance the reading experience in e-books, such as the enhanced e-books technology [18], which integrates multimedia features such as animation, sound effects, and illuminated text. As another example, Guedes [19] developed the Multisensorial Books model. This model adopts multisensorial media effects to enhance the quality of experience (QoE) of electronic books readers. It is also important to emphasize the audio-augmented

e-books [20], which combines audiobooks with e-books to transform the reading task into an audio-visual experience.

Although it is possible to find such studies, little is found about the development of models concerned in expanding the audiobooks' concept. Regarding this point, it is possible to emphasize two main studies. Interactive audiobooks [21] is one of these studies, where the authors combine the narratives with interactive elements from computer games. Interactive audiobooks have already received some improvements and modifications, in addition to the first implementation of this proposal [22]. The second main study presents an extension of audiobooks listening with a surrounding lighting system that changes its color based on emotional expression present in audiobooks passages [23]. No other approach related to the extension of audiobooks concepts has been found in the literature.

Considering the elements presented so far, the motivating question underlying this study's development is: 'how can we create multisensorial media technology that assists in the learning task and contemplates individuals with visual disabilities?'. The first point to consider here is the reason to use multisensorial media. In addition to improving the QoE of users, the benefits of multisensorial information on learning and memory have been fully discussed in cognitive sciences and psychology literature [24], [25].

The second point to highlight is that many multisensorial media approaches are not accessible to visually impaired individuals. We can refer to a large number of studies focused on presenting videos and multisensorial effects [26]–[29]. We can also refer to the Multisensorial Books model since it, by definition, does not consider its use by visually impaired individuals. It is important to note that another group of individuals could be benefited from this model: dyslexic individuals. The benefits of audiobooks use by dyslexia students are discussed by Selikowitz [30]. The author also states that audiobooks can circumvent reading difficulties. As another example, Milani et al. [31] concluded that audiobooks could significantly improve reading accuracy, school performance, and could also be associated with greater involvement in school activities.

In this scenario, this study proposes a new audiobook model, capable of providing multisensorial effects to audiobook listeners. This model is named as 'Multisensorial Audiobooks'. To develop our model, we considered three main points:

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- (i) the extension of Multisensorial Books concepts, since it can enhance the QoE of readers and have presented promising results [19], [32];
- (ii) the creation of audiobook technologies for impairment people (most audiobooks were initially introduced as accessibility tools [33]);
- (iii) the application of Multiple Intelligence concepts in education [34], since the literature indicates it can bring several benefits in the educational process [35]–[37].

Moreover, we developed the first application based on Multisensorial Audiobooks, named as MulSeBook. We also created the first story to be played in MulSeBook. This story was conceived based on Multiple Intelligence concepts. Initial experiments indicate MulSeBook can produce multisensorial media effects while playing stories.

The remaining of this work is divided as follows: in Section II, we briefly review some important work related to this research. In Section III, we present the Multisensorial Audiobooks model. In Section IV, we present the first prototype based on the Multisensorial Audiobooks model. In Section V, we present initial stage experiments. Finally, in Section VI, we discuss the contribution of this work and the next steps in this research.

II. RELATED WORK

Interactive audiobooks [21], [22] consists of a model to enhance audiobooks with interactive elements from computer games. The listener goes from having a passive experience to having an interactive experience with the narrative. The creation method uses story trees to create a non-linear story design, like the methods used by game designers. The proposed system is divided into two parts: (i) an authoring environment; (ii) a runtime system. Until the compilation of this paper, no other study, no experiments with Interactive audiobooks have been found in the literature.

N'Sink (Naturally-Spoken Ink) [20] consists of a system with an alignment algorithm and a prototype user interface to enhance e-books with its associated audiobook. The goal of N'Sink is to enable users to listen to the book at the same time they are reading. The system applies Text-to-Speech and audio alignment algorithms to enhance the readers' experience. Experimental results with 12 five-minute excerpts of 6 different audiobooks presented word alignment errors below 120ms for 90% of the words. The authors also highlight the possibility of personalized audiobooks creation, where one could create audiobooks with their own voice.

Multisensorial books [19] is a model that incorporates the use of Multiple Sensorial Media (MulSeMedia) elements (e.g., wind, smells) to enhance the QoE of readers. The proposed model consists of a reading device, a central platform to centralize the effects' execution and a set of MulSeMedia devices. The author also created a prototype to evaluate the conception of the model. The prototype was tested on a Galaxy S7 smartphone, and it was connected to a MulSeMedia device control platform via Bluetooth. This platform was implemented on a Raspberry Pi 3 board, and three effects

were configured: auditory, using an iHome connected to the Raspberry with an audio jack; olfactory, adopting a simple scent diffuser with jasmine scent and haptic, using a case fan to generate an air-flow. The initial results of this study indicate promising results. The author states that Multisensorial books can open a way to develop and improve new technologies for MulSeMedia effects in reading.

Inspired on Multisensorial books concepts, Vieira et al. [32] developed an empirical study to evaluate the reading of e-books and Multisensorial effects. The authors created a prototype capable of synchronizing sensory effects with the reading content. After that, they carried out experiments with a medieval action story, created with audio, wind, and light effects. Experiments were conducted with 14 subjects divided randomly into two groups: the control group and the experimental group. The results indicated that, at some story excerpts, MulSeMedia content could influence the ambient perception.

Recently, Kreitlon et al. [38] developed the Affective Multisensorial Books Model, based on e-books reading, multisensorial media (MulSeMedia) concepts, and affective computing area. This model proposes acquiring affective states from users to produce multisensorial media effects. It is motivated by the fact that MulSeMedia can improve individuals' quality of experience while affective computing can capture individuals' affective states during reading (e.g., boredom, drowsiness). Moreover, the study evaluates if it is possible to detect changes in individuals' affective states when reading. Galvanic Skin Response (GSR) electrodes were connected with the subject's index finger and middle finger in the experiments. Each of ten subjects read two different genres of texts: a poem and a water fountain manual. Results indicate a statistically significant difference in GSR signal.

In Orloff et al. [23], the authors present SentiBooks, a smartphone application to enhance audiobooks listening. This application aims to combine a surrounding lighting system to create an illumination that is in line with the emotional expression present in audiobooks passages. The authors adopted a Philips Hue Light Bulb to modify the lighting in the room. They assume that illumination influences our behavior and activities, such as our state of relaxation and our attention. Following these concepts, SentiBooks uses the OpenVokaturi library [39] to compute five emotional states in the voice: neutrality, happiness, sadness, anger, and fear. Next, considering these emotional states, SentiBooks changes the bulbs' light color with the following pattern: *neutral* emotion turns the light to white color; *anger* to red; *fear* to purple; *sadness* to blue; *happiness* to orange. The study also then evaluated SentiBooks usability with six persons. Results indicate that users can imagine themselves using the app; however, they stated that emotion recognition seems arbitrary at times. However, the authors plan to improve the emotion recognition module by exploring alternative software packages' performance. They also plan to combine the proposed approach with speech-to-text and textual sentiment analysis to perform multimodal emotion analysis.

The literature contains little work related to the enhancing of audiobooks. On the other hand, some studies were found concentrated on enhancing ebooks reading with MulSeMedia. However, there are only two works focused on enhancing audiobooks with multimedia effects. With these considerations, this work proposes the Multisensorial Books model and develops an application, based on this model, named Mulsebooks. It is important to note that SentiBooks (proposed by Ortloff et al., as mentioned previously) technically do not use MulSeMedia effects, since it presents only one multimedia effect (i.e., modifies the lighting in the room). Moreover, it is not inclusive to blind individuals since they cannot see the color lighting changes. We can emphasize our work's main contribution: to create a new audiobook model and application, accessible to visually impaired individuals, and capable of providing multisensorial effects to audiobook listeners.

III. MULTISENSORIAL AUDIOBOOKS

Multisensorial media refer to the combination of traditional media (e.g., text, graphical images) with multiple sensory effects so that other senses are stimulated [40], [41]. As already discussed, these effects can improve the QoE of multisensorial book readers. In this scenario, the main goal of the Multisensorial audiobooks model is to enable multisensory effects in audiobooks. It is important to note that, although the objective of this study is to design a model that serves people with visual impairments, the proposed model is not exclusively dedicated to this objective.

Figure 1 illustrates the Multisensorial audiobooks model. The user access a reading device (e.g., tablet, cell phone) to hear a story. The device communicates with a control platform, which triggers the effects related to the story. The effects can be visual, auditory, haptic, gustatory, or olfactory.

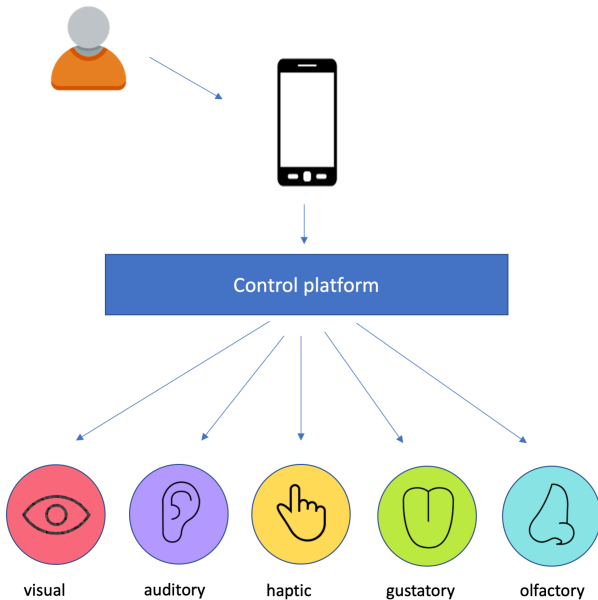


Fig. 1: Multisensorial audiobooks model.

There are several works in the literature focused on developing devices to provide multisensorial effects. These effects can activate one or more human senses. We can highlight some studies found in the literature concerning each of the human senses: (i) visual, RGB smart LED lights can change the ambient lighting [42]; (ii) auditory, ambient sounds, including noise and/or music, can be played in the ambient (e.g., birds, office noises, and traffic) [43] (iii) haptic, a fan/ventilator can be used to address haptic sensations [42]; (iv) gustatory, electrical pulses on the tongue through silver electrodes can be used to address gustatory sensations; (v) olfactory, where fans can be used to disperse scents and stimulate the olfactory system [44].

IV. MULSEBOOK

This section presents the first prototype of Multisensorial audiobooks model, a system named MulseBook. Section IV-A discusses the input files read by MulseBook. Section IV-C presents the interface provided. Section IV-B discusses the architecture built. Section IV-D presents the first story conceived for MulseBook.

A. Software Development

MulseBook was developed in Unity 2019.3.5. It starts by reading an XML file containing audiobook information, instructions about what effects will be considered, and when these effects will be triggered. Fig. 2 illustrates an example of XML to be read in MulseBook. The tag *AudioBook* contains three attributes: *id*, which represents a unique id for each audiobook; a *title*, which represents the title of the audiobook; an *audio* file, with an audio representation of the audiobook title.

The tag *AudioBook* also encompasses a set of *Signal* tags. Each *Signal* tag can have four attributes:

- device, which represents the device label (e.g., speaker, diffuser).
- state, which describes the action to be executed, that is, to turn the device on or off.
- property, which is acceptable in some tags; it represents the path of the sound file, in the case of a speaker;
- time, which represents the moment, after the beginning of the story, that the device will be turned on or off.

```
<AudioBook id="1" title="Cida e Adão numa experiência de
leitura multisensorial" audio="audiol.wav">
  <Signal device="fan" state="on" time="00:00:29"></Signal>
  <Signal device="fan" state="off" time="00:00:40"></Signal>
  <Signal device="speaker" state="on" property="rain.wav"
    time="00:00:43"></Signal>
  <Signal device="speaker" state="off" property="rain.wav"
    time="00:00:48"></Signal>
  <Signal device="diffuser" state="on" time="00:01:06"></Signal>
  <Signal device="diffuser" state="off" time="00:01:12"></Signal>
</AudioBook>
```

Fig. 2: Example of XML file in MulseBook.

B. Hardware Architecture

In this study, the architecture designed to work with MulseBook is composed of: (i) a computer as the *device* to access the audiobooks, in this case, a desktop PC equipped with an Intel Core i7 processor, 8mb RAM, and Windows 10; (ii) an ESP8266 module (a low-cost Wi-Fi chip with full TCP/IP stack) to represent the *control platform*; (iii) three devices used to activate auditory, haptic, and olfactory effects. The devices are described below.

- auditory, a JBL GO Bluetooth Speaker.
- olfactory, a 400ml aroma diffuser with infrared remote control.
- haptic, a 45-watt Mondial fan adapted with an infrared circuit to turn it on and off.

MulseBook communicates with the audio speaker using Bluetooth technology. The communication between MulseBook and ESP8266 is conducted using the Wi-Fi. Then, the ESP8266 communicates with the diffuser and with the fan by using two distinct infrared circuits.

C. Interface

The interface of MulseBook was designed to be easily learned and accessible for visually impaired people. When the user executes the application, the loading screen exhibits the application's name, as illustrated in 3a. Next, the MulseBook shows the first audiobook available in the library: the audiobook title and an audio representation of the title, as shown in Fig. 3b. Users can browse the audiobook titles by dragging their fingers across the screen, and the application exhibits information about the current audiobook. At the same time, audio information is displayed containing the name of the audiobook. This process is illustrated in Fig. 3c.

When the user chooses an audiobook and presses the screen, MulseBook changes to another screen (Fig. 3d) and starts by playing an audio representation of the title. This screen presents two buttons. The first is used to return the user to the previous screen. It is located at the bottom of the screen. The second button can be used to play and pause the audiobook playing. It is located in the center of the screen. It is important to note that all inside interfaces of MulseBook present a help button, located at the top-right corner of the screen.

D. The First Story

The first story produced for MulseBook was developed inspired by the theory of multiple intelligences. It was motivated by the importance of learning naturalist intelligence concepts [45]. Moreover, there was a decision to make this first story also dedicated to visually impaired people. For this reason, no visual effect was used (e.g., lights changing in the room).

The story starts with three characters reading a book about a naturalistic scene. Three effects are configured, as described in the XML file presented in Fig. 2: a fan, used to represent the wind; a sound of rain, played to simulate the rain falling; and a diffuser, prepared with a scent of flowers, to represent the word 'perfumam' (perfumed).

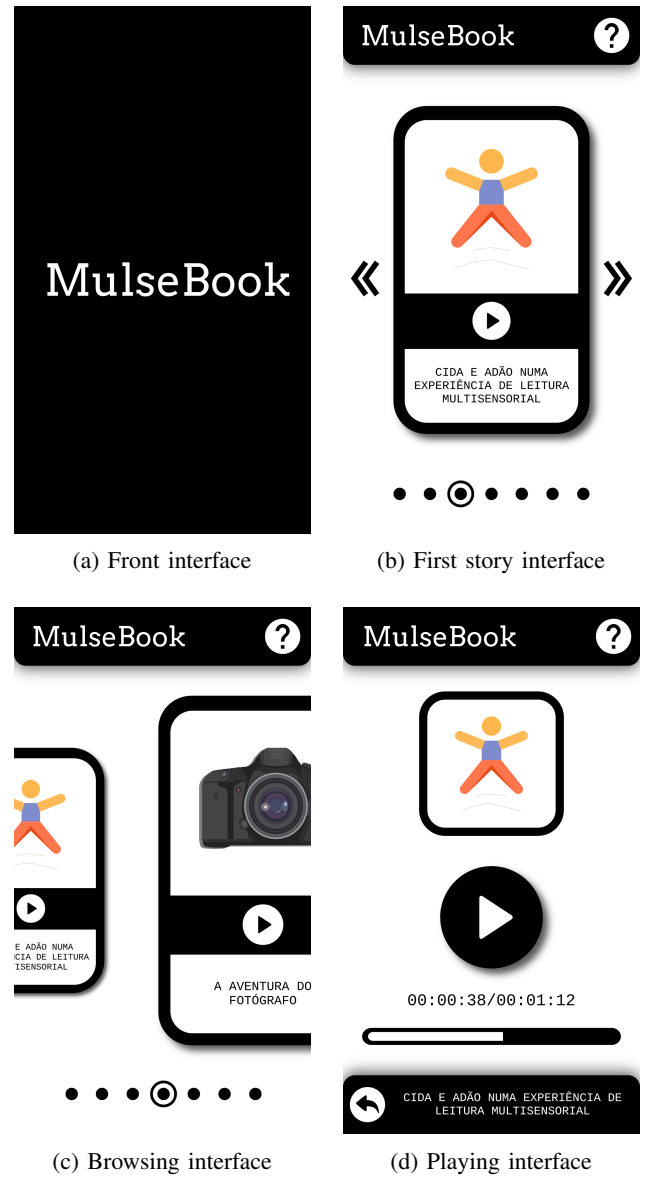


Fig. 3: MulseBook interfaces. Figure 3(a) presents the front interface; Figure 3(b) shows the interface of the first story proposed to be executed in MulseBook application. Figure 3(c) illustrates the browsing interface. Finally, Figure 3(d) presents MulseBook playing interface.

Figure 4 presents a passage extracted from the first created multisensorial audiobook, which is *Cida e Adão numa leitura multissensorial* (Cida and Adão in a multisensorial reading, in English). In this passage, the words presented in blue color represent words related to multimedia effects. For example, the word 'soprar' (blows), which represents the wind that blows, is associated with the fan to simulate the wind. The words 'ondas' (waves) triggers a sound of waves, and the word 'perfumaram' (perfumed) turns the diffuser on (with a scent of flowers). Finally, the phrase 'chuva rega a terra' (rain waters the soil) triggers a sound of rain in the soil.

Que belo é o mundo,
 Com tantos bichinhos...
 O céu estrelado,
 O vento a *soprar*;
 O sol bem quentinho,
 Nuvens brancas bailando no ar;
 As *ondas* do mar
 Brincando na praia
 Pra lá e pra cá...
 Flores do campo
Perfumam a terra
 Rosas, lírios, jasmims!
 E para fazer a semente brotar
 A *chuva rega a terra*,
 Fazendo a planta crescer, crescer e crescer...
 Da Natureza colhemos legumes, verduras,
 Maças, uvas, bananas, frutas maduras
 E tantas coisas gostosas pra gente comer!

Fig. 4: Passage extracted from the audiobook *Cida e Adão numa leitura multissensorial*. The words in blue represent the points where the multimedia effects are executed.

V. PRELIMINARY RESULTS

The conception of MulseBook is still a work in progress. However, we did initial experiments to validate if MulseBook can produce multisensorial media effects while playing a story. This evaluation is essential since it can indicate future directions for this work.

All MulseBook interface transitions worked as expected. The connection between the computer and the three multimedia devices (i.e., Bluetooth Speaker, aroma diffuser and the fan) has also been validated. The application could read the XML file and trigger the effects at the moment configured in the XML.

VI. DISCUSSION AND FUTURE WORK

This work first discussed the relevance of multisensorial effects to Besides, based on literature studies, it highlighted that multisensorial information is an important key to learning and memory. Moreover, this study emphasized the benefits of Multiple Intelligence concepts in the educational process. These theoretical foundations motivated the construction of this work.

The importance of the accessibility of multisensorial technology by impaired people was also presented. In addition to visually impaired individuals, this study can benefit individuals with dyslexia. With these considerations, this study's main goal was to develop multisensorial technology that could benefit visually impaired and dyslexia individuals.

To achieve the primary goal, we proposed a Multisensorial Audiobook model. The core idea behind this model is to extend audiobooks with multisensorial approaches since multisensorial information can benefit learning, memory, and Quality of Experience of users. Moreover, we created the

first prototype of Multisensorial Audiobook (MulseBook) and the first story to be played by this prototype, named *Cida e Adão numa leitura multissensorial* (Cida and Adão in a multisensorial reading). Since this is still a work in progress, we evaluated the dynamics of the application functioning. Initial results suggest this research is moving in the right direction.

The next step of this research is to evaluate MulseBook application usability with students of the Federal Center of Technological Education of Rio de Janeiro (CEFET/RJ). In the second stage, we aim to evaluate MulseBook with visually impaired individuals, partnering with institutions that support blind people. Finally, we pretend to evaluate MulseBook's use by dyslexic individuals.

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